Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CS5AI02]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

BIG DATA MANAGEMENT

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Define big data. Describe the 5 V's of Big data which separates big data from traditional datasets.	**07**
	b.	Summarize structured, semi-structured and unstructured data types with an Example of each.	**07**
2.	**a.**	Explain Hadoop Architecture and its components with proper diagram.	**07**
	b.	Write a word counting program to count the words using Apache Spark model. Explain each components used in the program.	**07**
3.	**a.**	Illustrate read and write operation in HDFS with a suitable diagram.	**08**
	b.	How files are stored in HDFS? Explain how HDFS is different from Unix file system.	**06**
		OR	
4.	**a.**	Describe the following concepts: **i.** FS Image **ii.** Edit-logs **iii.** Check-pointing	**06**
	b.	Name and illustrate the commands for starting and stopping different Hadoop daemon processes for HDFS and MapReduce in a Hadoop cluster.	**08**
5.	**a.**	Explain when not to use MapReduce. Describe map side and reduce side join.	**07**
	b.	Why do you use combiner and partition function in MapReduce while it has Mappers and Reducers? Explain in detail.	**07**
		OR	
6.	**a.**	Explain three input and output format types of MapReduce.	**08**
	b.	Explain the use of mapper and reducer in a MapReduce program in obtaining efficiency than the traditional programs in the context of classical WordCount program.	**06**
7.	**a.**	Draw YARN architecture and explain the YARN Components.	**06**
	b.	Write down a short note on different modes of Hadoop installation and execution.	**08**
		OR	

8. a. Explain the differences between Hadoop 1.x and 2.x Architecture. Compare and Contrast Hadoop with Relational Databases. **07**

b. List the steps of the execution of Job in Yarn Cluster. Explain the details of the following in cluster setup and installation: **07**

i. Creating Hadoop user

ii. SSH Configuration